

ECEN 454 - Lab Report

Lab Number: 9

Lab Title: Optimization using Logical Effort

Section Number: 341

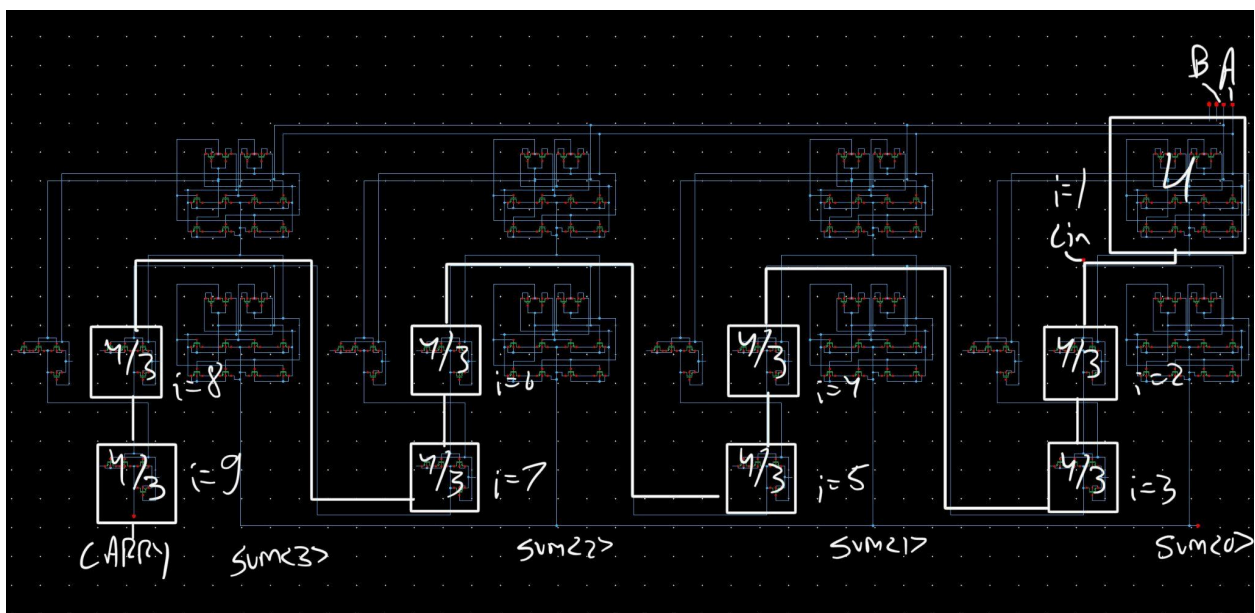
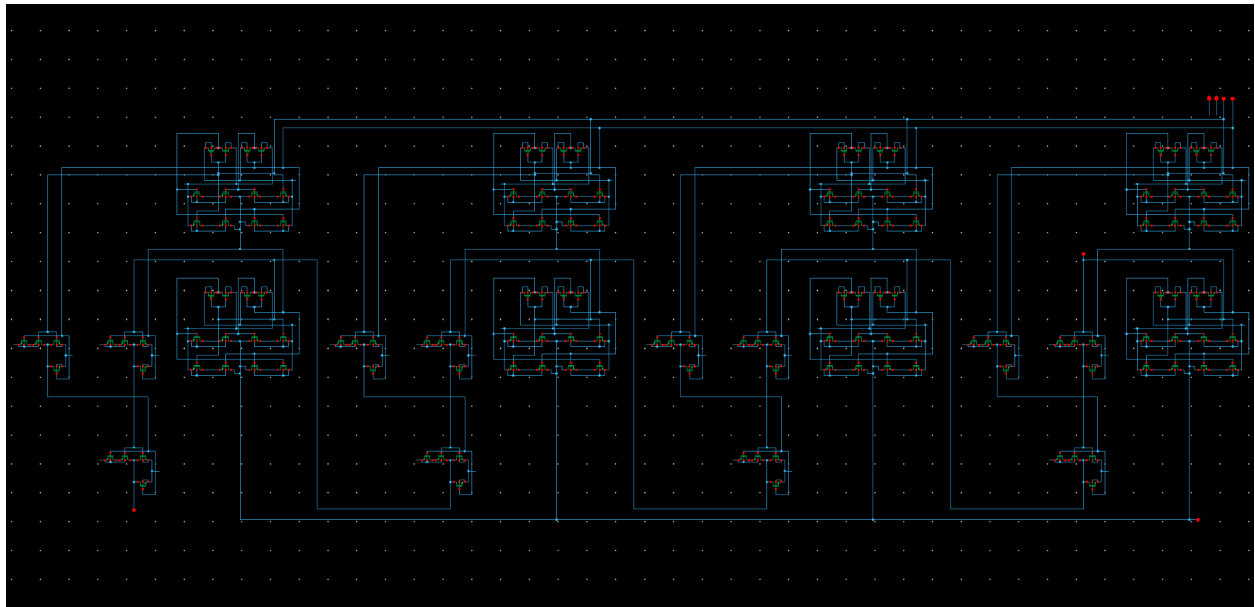
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Date: 11/17/2024

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Circuit Schematic w/ Critical Path Highlighted



Logical Effort Calculations (G, B, H) & Procedure

Explanation of Procedure

I used Google Sheets for all logical effort calculations, following the procedure as laid out in the lab manual and the provided Sutherland_Ch1.pdf. [I have shared this google sheet at this link in case there are any questions about my calculations.](#) My procedure and formulas were:

I first looked up the values of sink capacitances from Lab 3. This gave me:

$$C_{\text{NAND}} = 2.23 \text{ fF}$$

$$C_{\text{XOR}} = 7.51 \text{ fF}$$

I then calculated the Path Logical Effort (G) by taking the product of each of the gate's g:

$$G = 4 * (4/3)^8 = 39.955$$

I then calculated Path Electrical Effort (H) as $C_{\text{load}} / C_{\text{XOR}}$:

$$H = C_{\text{load}} / C_{\text{XOR}} = 30 \text{ fF} / 7.51 \text{ fF} = 3.995$$

Finally, I calculated Path Branching Effort (B) as the product of $b_0 * b_1 * \dots * b_9$. I calculated b_i for each device using the formulas given in the lab manual:

$$b_1 = (C_{\text{XOR}} + C_{\text{NAND}}) / C_{\text{XOR}} = 1.30$$

$$b_{3, 5, 7, 9} = 1$$

$$b_{2, 4, 6, 8} = (C_{\text{XOR}} + C_{\text{NAND}}) / C_{\text{NAND}} = 4.37$$

$$B = b_1 * b_2 * \dots * b_9 = 1.30 * 1 * (4.37)^4 = 471.99$$

Finally, I calculated Path Effort (F) as $F = G * B * H$, and Stage Effort ($\hat{f}$) as $\hat{f} = F^{1/9}$

Final Values

Cnand	2.23E-15
Cxor	7.51E-15
Gxor	4
Gnand	1.333333333
G	39.95488493
H	3.99
B	471.9928233
F	75333.23146
$\hat{f}$	3.482470127

Resizing Procedure & Final Transistor Sizes

Using the procedure above, I found a value of $\hat{f} = 3.482$. Since If $2.7 < \hat{f} < 4$, I performed the following resizing:

1. For the i-th gate ($i = 9, 8, \dots, 1$), calculate $C_{in,i} = g_i C_{out,i} / \hat{f}$
2. If $C_{in,i} < C_{gate}$ keep the size. Otherwise, resize the width of W_n and W_p by ratio $C_{in,i} / C_{gate}$

To do this, I first calculated $C_{in,i}$ for each gate starting from $i = 9$ (final NAND gate, as labeled in my critical path schematic) and ending at $i = 1$ (first XOR gate). The gates at $i = 7, 5, 3$, and 1 are highlighted as they have an additional capacitance for the XOR gate that needs to be added. This is reflected in the formulas given below:

Gate (i)	Formula	Cin	Resize by (if green)
9 (NAND)	$C_{in,9} = (4/3 * C_{load}) / \hat{f}$	1.15E-14	5.150717473
8 (NAND)	$C_{in,8} = (4/3 * C_{in,9}) / \hat{f}$	4.40E-15	1.972055193
7 (NAND)	$C_{in,7} = [4/3 * (C_{in,8} + C_{XOR})] / \hat{f}$	4.56E-15	2.044437043
6 (NAND)	$C_{in,6} = (4/3 * C_{in,7}) / \hat{f}$	1.75E-15	0.7827536083
5 (NAND)	$C_{in,5} = [4/3 * (C_{in,6} + C_{XOR})] / \hat{f}$	3.54E-15	1.589089147
4 (NAND)	$C_{in,4} = (4/3 * C_{in,5}) / \hat{f}$	1.36E-15	0.6084145598
3 (NAND)	$C_{in,3} = [4/3 * (C_{in,4} + C_{XOR})] / \hat{f}$	3.39E-15	1.522339956
2 (NAND)	$C_{in,2} = (4/3 * C_{in,3}) / \hat{f}$	1.30E-15	0.5828582971
1 (XOR)	$C_{in,1} = [4 * (C_{in,2} + C_{XOR})] / \hat{f}$	1.01E-14	1.347402728

I then calculated the resizing factor for each gate as $C_{in,i} / C_{gate}$ in the “Resize by” column. If the value is less than one, then we get $C_{in,i} < C_{gate}$ and keep the size. I used conditional formatting to highlight all gates with a value greater than one and only these were resized.

Below are the sizing used for unoptimized and optimized gates:

Unoptimized Gate Sizing:

Unoptimized		
Inv	W_p	1100n
	W_n	400n
NAND	W_p	600n
	W_n	300n
XOR	W_p	1600n
	W_n	400n

Optimized Gate Sizing:

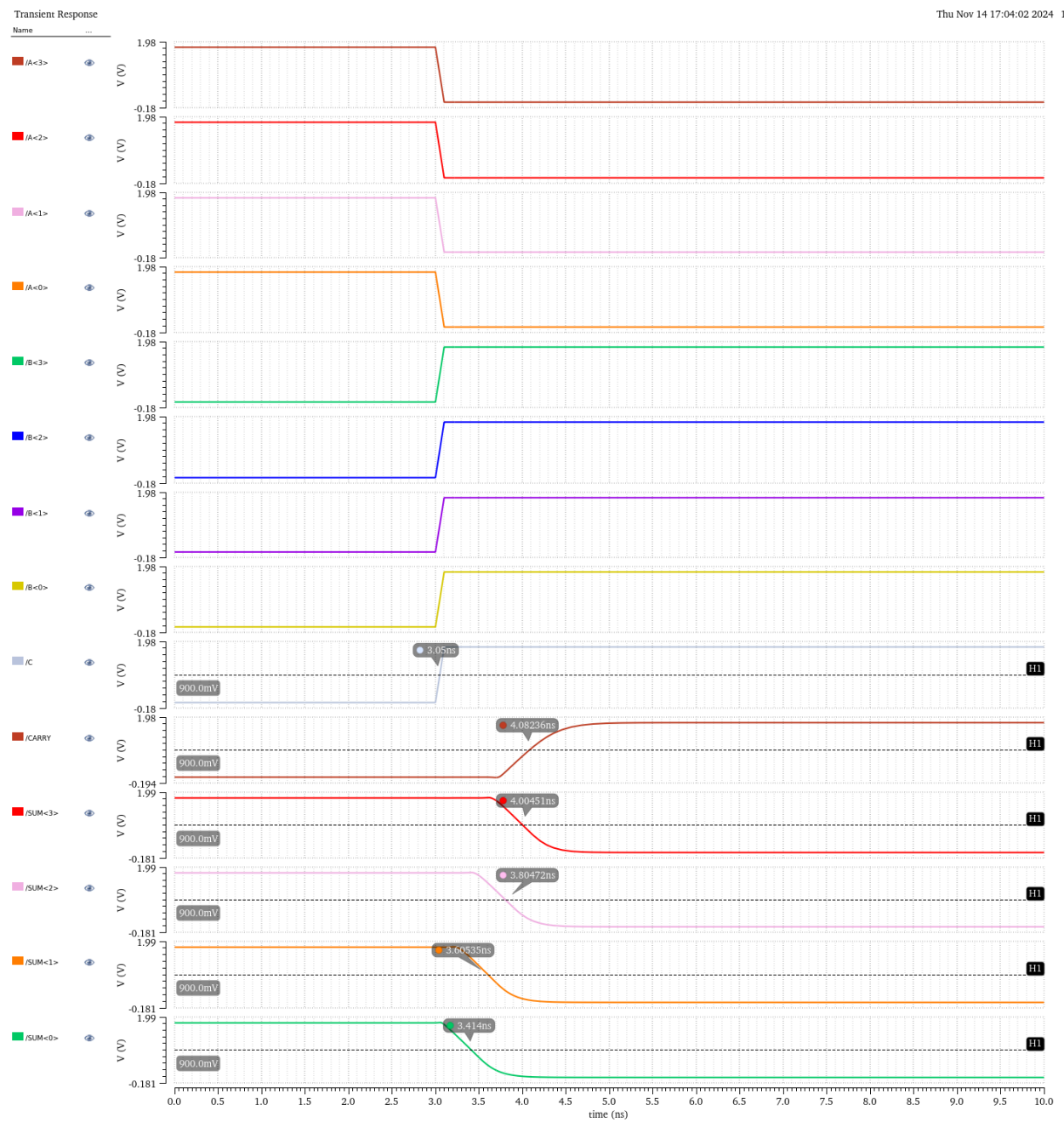
Note: The gate numbers given here correspond to the gates on the critical path schematic (using the i variable). For the “optimized” sizing, these specific sizes were used for each gate on the critical path. For all gates NOT on the critical path, the sizing was left unchanged.

Optimized		
NAND9	W_p	3090.430484
	W_n	1545.215242
NAND8	W_p	1183.233116
	W_n	591.6165578
NAND7	W_p	1226.662226
	W_n	613.331113
NAND6	W_p	600
	W_n	300
NAND5	W_p	953.4534882
	W_n	476.7267441
NAND4	W_p	600
	W_n	300
NAND3	W_p	913.4039735
	W_n	456.7019867
NAND2	W_p	600
	W_n	300
XOR1	W_p	2155.844364
	W_n	538.9610911
XOR1 (INV)	W_p	1482.143001
	W_n	538.9610911

Simulation Waveforms

Case 1: A=0000, B=1111, Carry In=1

Unoptimized

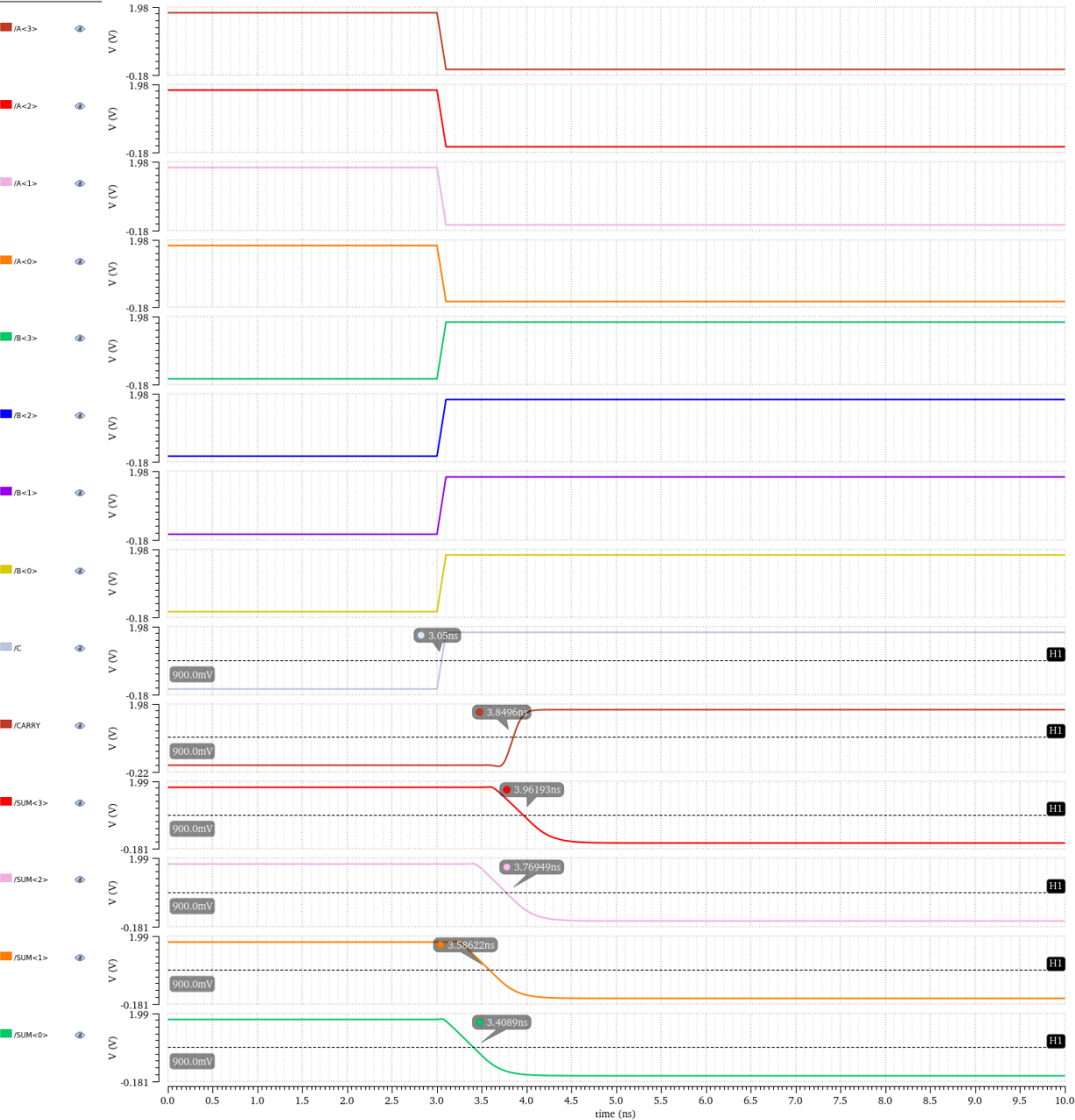


Optimized

Transient Response

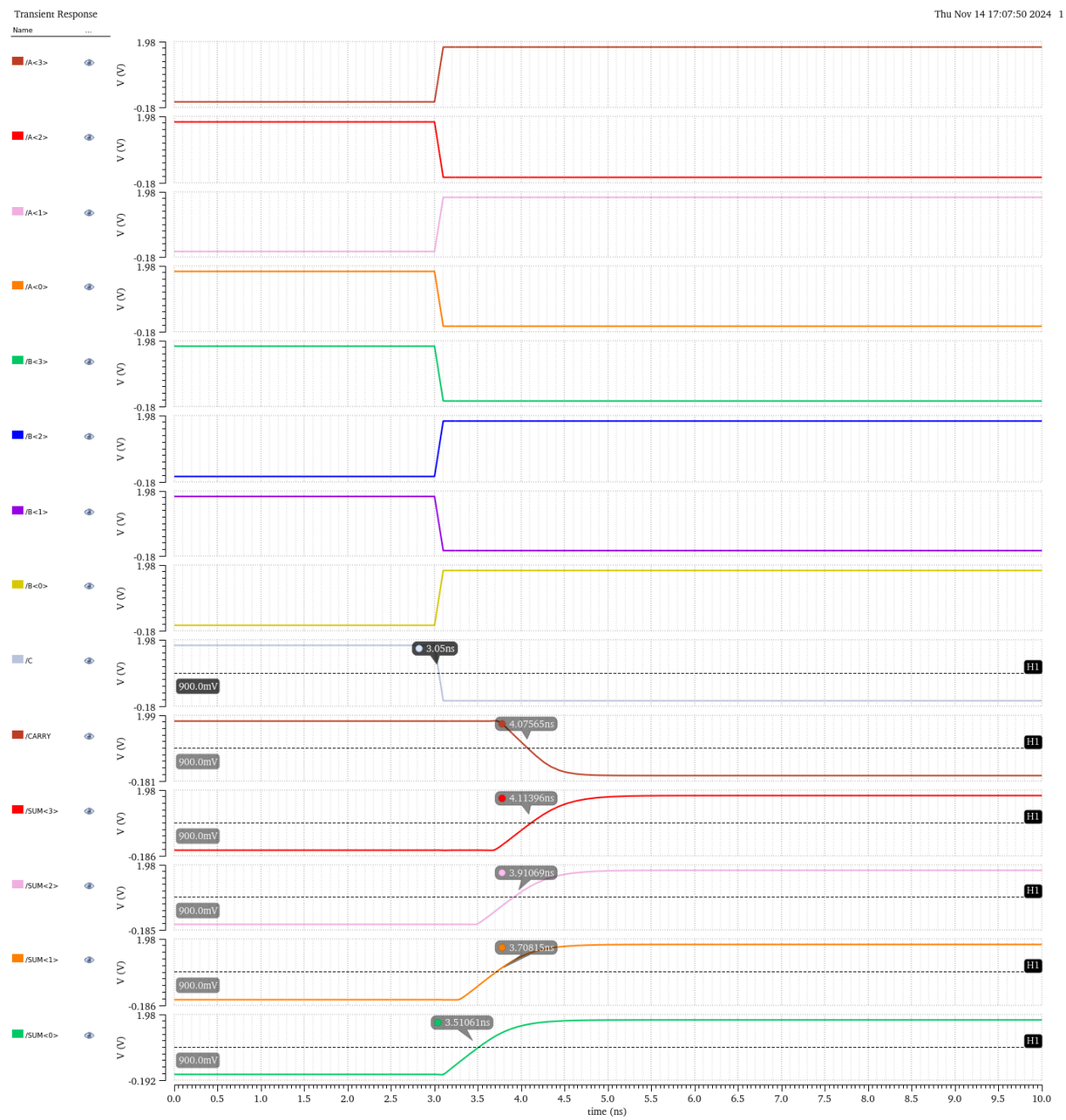
Thu Nov 14 17:46:42 2024 1

Name

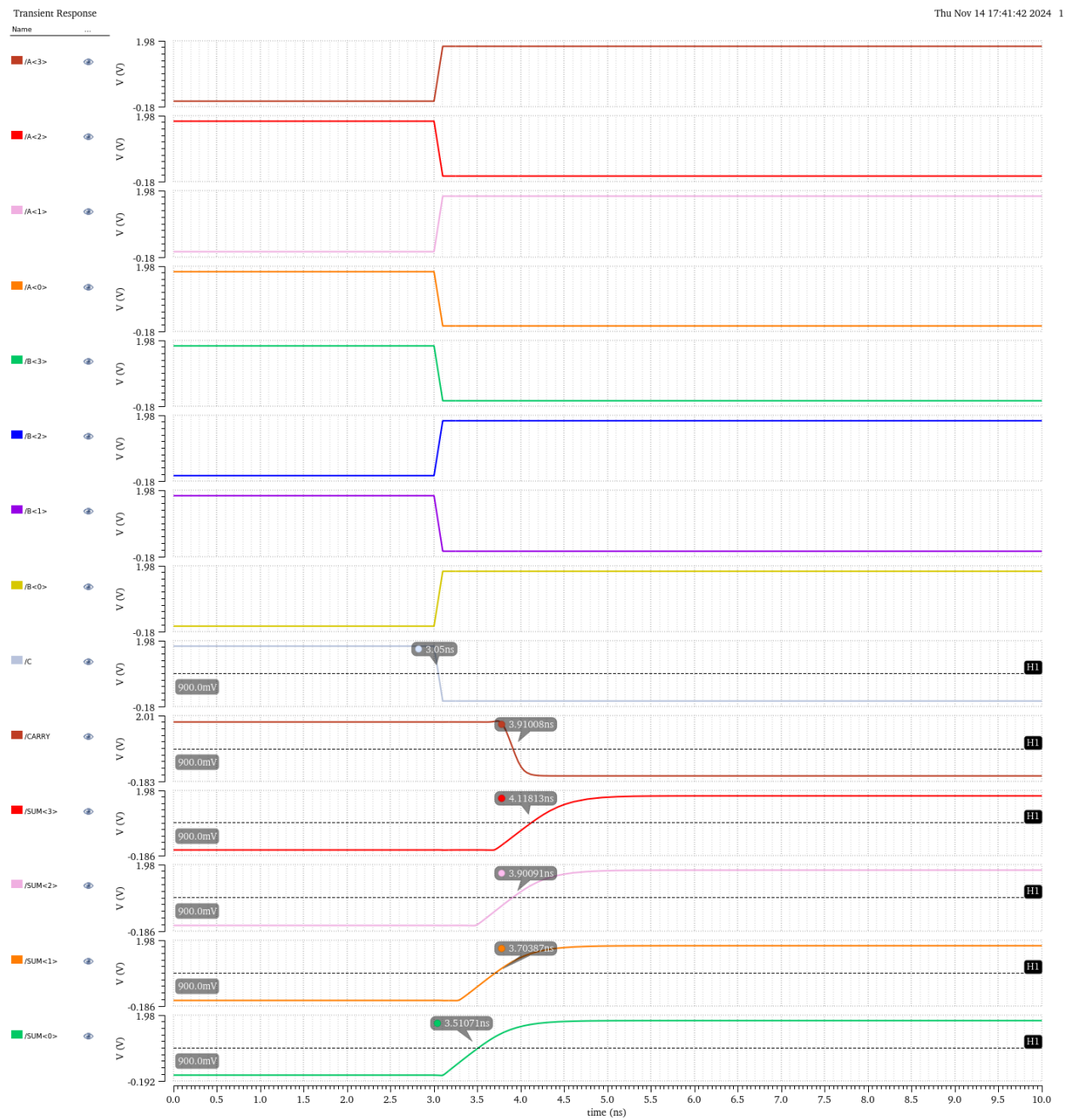


Case 2: A=1010, B=0101, Carry In=0

Unoptimized

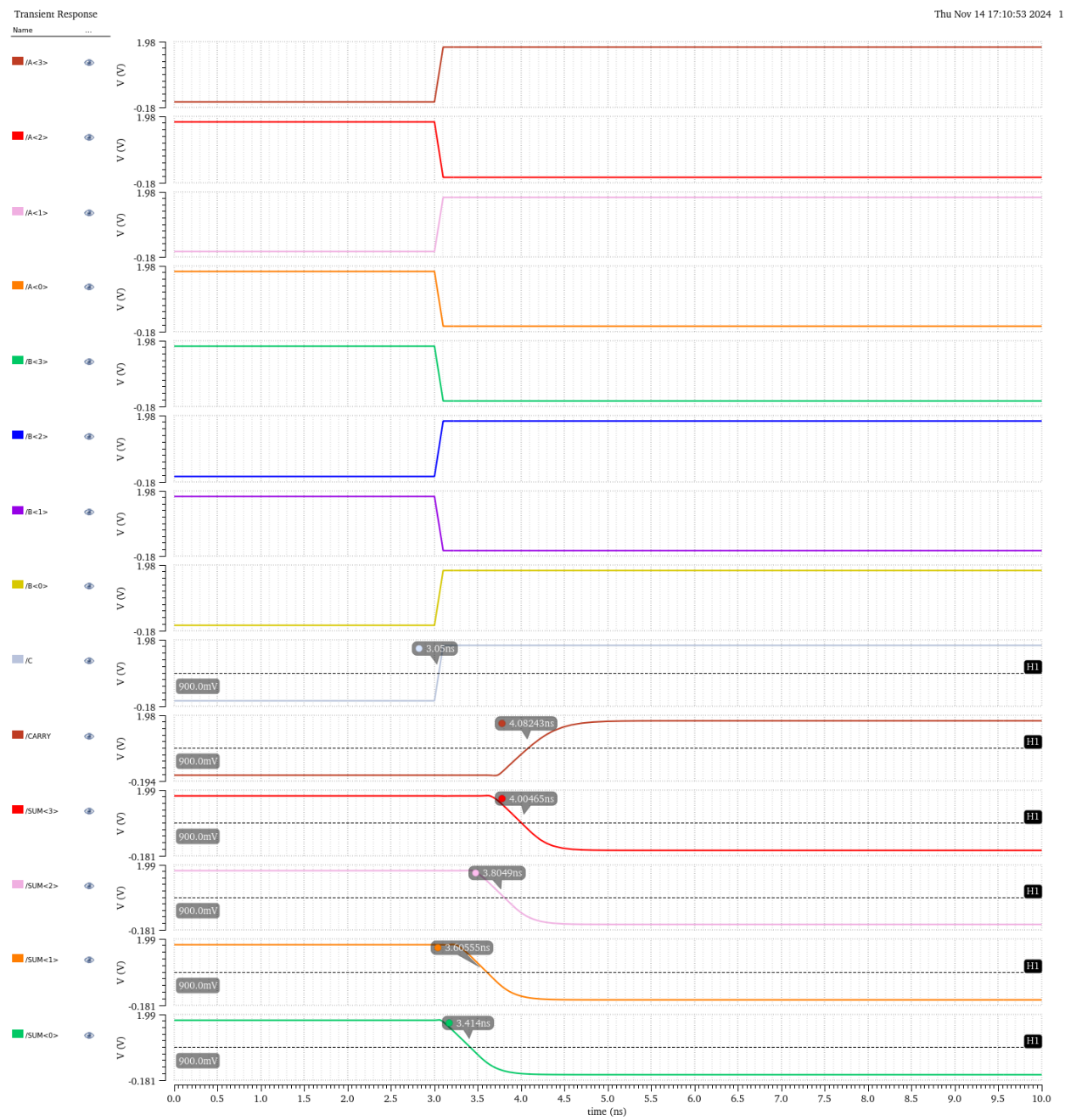


Optimized



Case 3: A=1010, B=0101, Carry In=1

Unoptimized

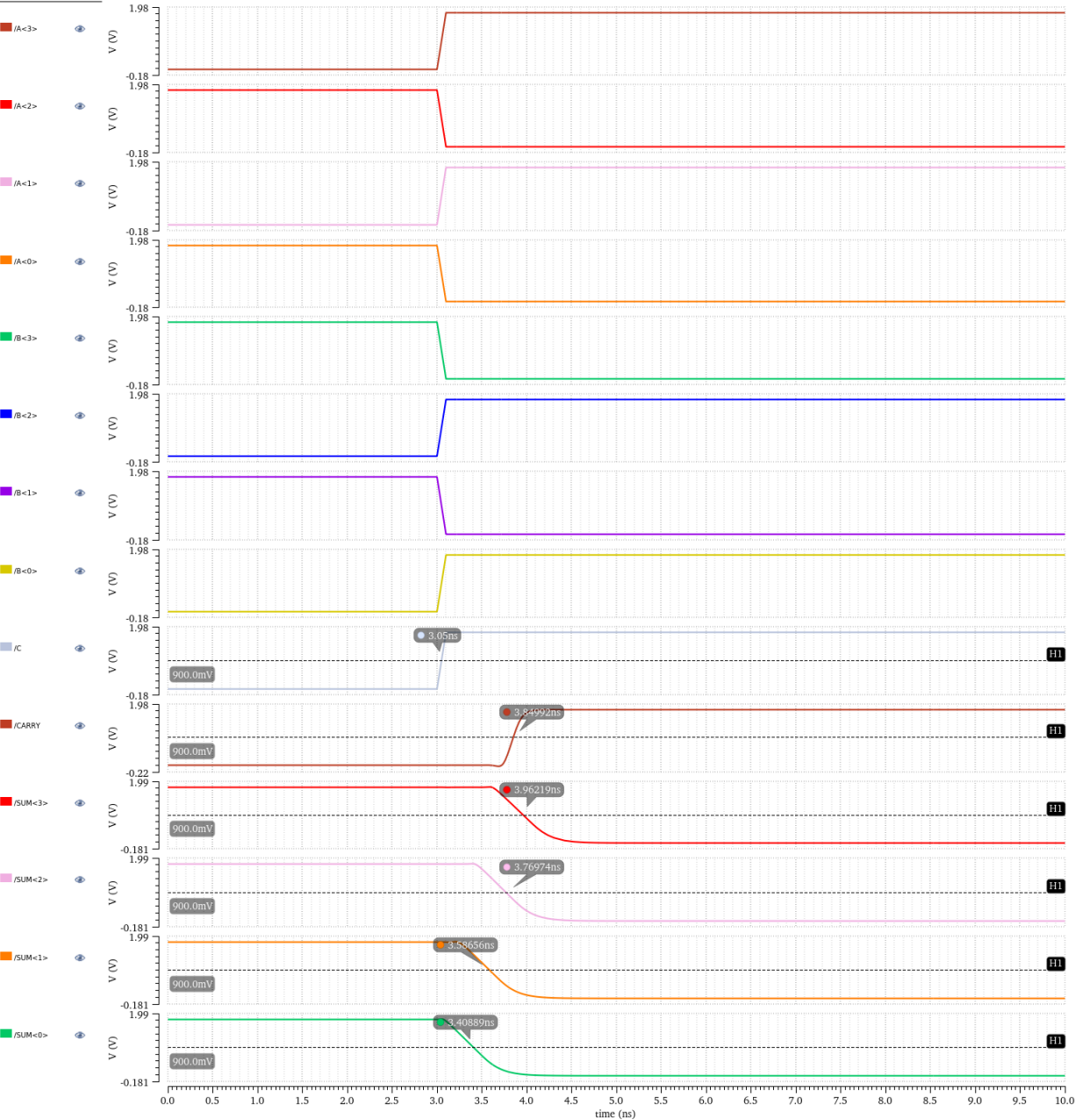


Optimized

Transient Response

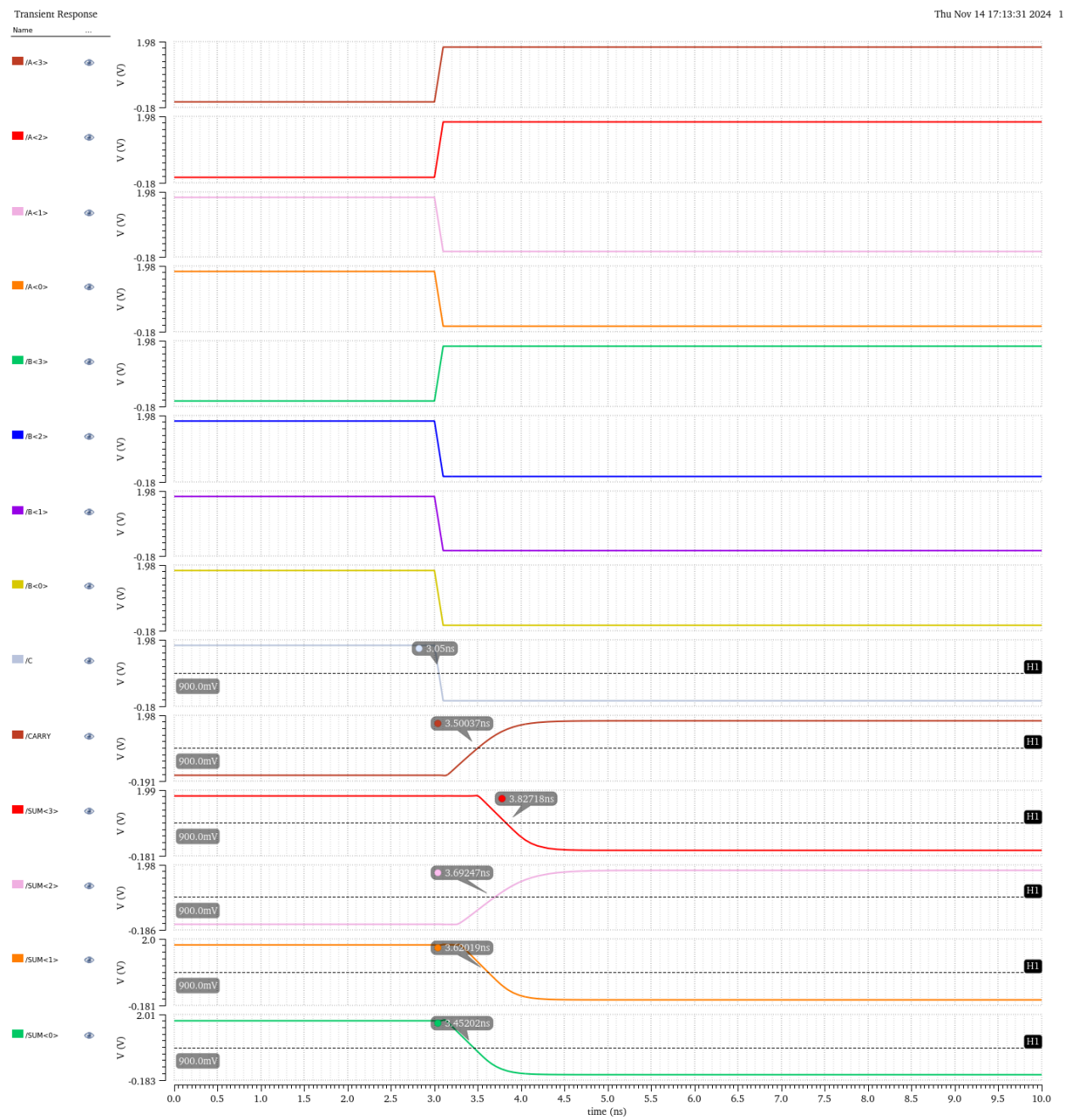
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Name



Case 4: A=1100, B=1000, Carry In=0

Unoptimized

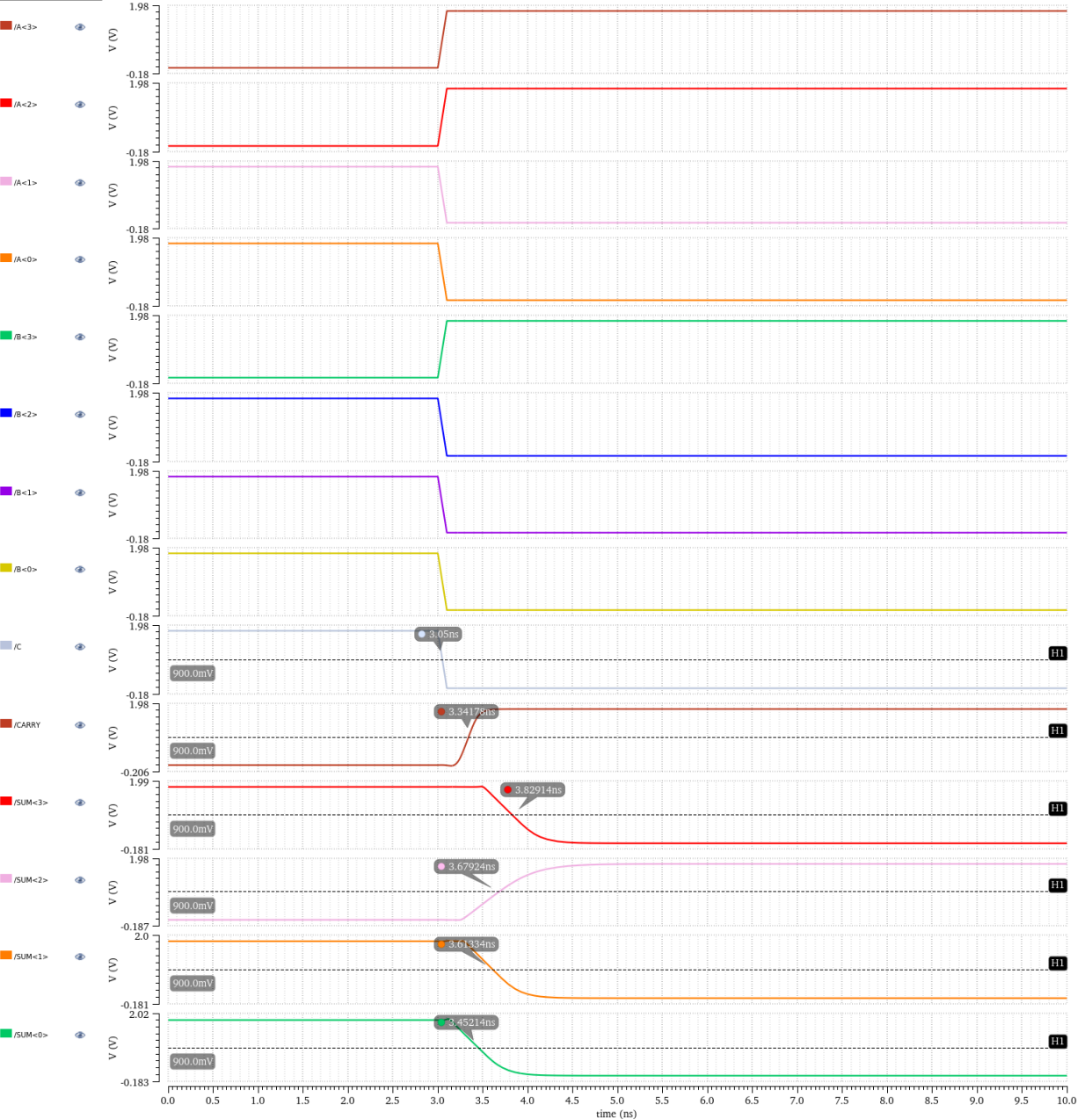


Optimized

Transient Response

Thu Nov 14 17:37:36 2024 1

Name



Delays, Unoptimized vs. Optimized

Note: I used conditional formatting here to color the cells and make any improvement in delay for the optimized design easier to spot. Delay is evaluated as Rise Time - 3.05, which was the measured time of the rising edge of the input signals (shown in the simulation waveforms).

Case	Pin	Rise Time (ns)		Delay (ps)	
		Non-Optimized	Optimized	Non-Optimized	Optimized
A=0000 B=1111 Carry In=1	CARRY	4.08236	3.8496	1032.36	799.6
	SUM<3>	4.00451	3.96193	954.51	911.93
	SUM<2>	3.80472	3.76949	754.72	719.49
	SUM<1>	3.60535	3.58622	555.35	536.22
	SUM<0>	3.414	3.4089	364	358.9
A=1010 B=0101 Carry In=0	CARRY	4.07565	3.91008	1025.65	860.08
	SUM<3>	4.11396	4.11813	1063.96	1068.13
	SUM<2>	3.91069	3.90091	860.69	850.91
	SUM<1>	3.70815	3.70387	658.15	653.87
	SUM<0>	3.51061	3.51071	460.61	460.71
A=1010 B=0101 Carry In=1	CARRY	4.08243	3.84992	1032.43	799.92
	SUM<3>	4.00465	3.96219	954.65	912.19
	SUM<2>	3.8049	3.76974	754.9	719.74
	SUM<1>	3.60555	3.58656	555.55	536.56
	SUM<0>	3.414	3.40889	364	358.89
A=1100 B=1000 Carry In=0	CARRY	3.50037	3.3417	450.37	291.7
	SUM<3>	3.82718	3.82914	777.18	779.14
	SUM<2>	3.69247	3.67924	642.47	629.24
	SUM<1>	3.62019	3.61334	570.19	563.34
	SUM<0>	3.45202	3.45214	402.02	402.14

Power Consumption, Unoptimized vs. Optimized

The optimized design consumed ~10 uW more power for all tests. This is expected, as the larger transistor sizing causes higher power consumption.

	Power	
Case	Non-Optimized	Optimized
A=0000 B=1111 Carry In=1	55.73 uW	63.74 uW
A=1010 B=0101 Carry In=0	105.5 uW	115.4 uW
A=1010 B=0101 Carry In=1	55.89 uW	63.89 uW
A=1100 B=1000 Carry In=0	81.59 uW	93.68 uW

Area, Unoptimized vs. Optimized

Here, the area is in units of picometers squared. Calculations are given in the Appendix below.

Area	
Non-Optimized	Optimized
20.19 pm ²	71.6 pm ²

Appendix: Area Calculations

I calculated the area by finding the area of each transistor type (XOR_{NMOS} , XOR_{PMOS} , etc.) and then multiplying this by the product of the number of transistors in each gate and the number of instances of each gate in the design. For the unoptimized design, I considered each of the critical path gates ($i = 9, 8, \dots, 1$) separately. I used the below table of formulas for the calculations.

Unoptimized						
Gate	Transistor	Width	Length	# of Gates	# Transistors (each)	Area
Inv	W_p	1.10E-06	2.00E-07	16	1	3.52E-12
	W_n	4.00E-07	2.00E-07	16	1	1.28E-12
NAND	W_p	6.00E-07	2.00E-07	12	2	2.88E-12
	W_n	3.00E-07	2.00E-07	12	2	1.44E-12
XOR	W_p	1.60E-06	2.00E-07	8	4	1.02E-11
	W_n	4.00E-07	2.00E-07	8	4	2.56E-12
						2.19E-11
Optimized						
Gate	Transistor	Width	Length	# of Gates	# Transistors (each)	Area
Inv	W_p	1.10E-06	2.00E-07	14	1	3.08E-12
	W_n	4.00E-07	2.00E-07	14	1	1.12E-12
NAND	W_p	6.00E-07	2.00E-07	4	2	9.60E-13
	W_n	3.00E-07	2.00E-07	4	2	4.80E-13
XOR	W_p	1.60E-06	2.00E-07	7	4	8.96E-12
	W_n	4.00E-07	2.00E-07	7	4	2.24E-12
NAND9	W_p	3.09E-06	2.00E-07	1	2	1.24E-12
	W_n	1.55E-06	2.00E-07	1	2	6.18E-13
NAND8	W_p	1.18E-06	2.00E-07	1	2	4.73E-13
	W_n	5.92E-07	2.00E-07	1	2	2.37E-13
NAND7	W_p	1.23E-06	2.00E-07	1	2	4.91E-13
	W_n	6.13E-07	2.00E-07	1	2	2.45E-13
NAND6	W_p	6.00E-07	2.00E-07	1	2	2.40E-13
	W_n	3.00E-07	2.00E-07	1	2	1.20E-13
NAND5	W_p	9.53E-07	2.00E-07	1	2	3.81E-13
	W_n	4.77E-07	2.00E-07	1	2	1.91E-13

NAND4	W_p	6.00E-07	2.00E-07	1	2	2.40E-13
	W_n	3.00E-07	2.00E-07	1	2	1.20E-13
NAND3	W_p	9.13E-07	2.00E-07	1	2	3.65E-13
	W_n	4.57E-07	2.00E-07	1	2	1.83E-13
NAND2	W_p	6.00E-07	2.00E-07	1	2	2.40E-13
	W_n	3.00E-07	2.00E-07	1	2	1.20E-13
XOR1	W_p	2.16E-06	2.00E-07	1	4	1.72E-12
	W_n	5.39E-07	2.00E-07	1	4	4.31E-13
XOR1 (INV)	W_p	1.48E-06	2.00E-07	2	4	2.37E-12
	W_n	5.39E-07	2.00E-07	2	4	8.62E-13
						7.16E-11